

What is your current role?

I am a software developer.

And how long have you worked with software development?

I think 4 years.

What AI tools do you use in your work for development?

Yeah, it has become very strong now recently, and there are some tools that I started using now and I am getting used to. But, in the beginning I used only ChatGPT and Gemini. They are the two that I use the most in my daily work. But now code development with an AI assistant has become very strong, both Cursor and Cloud, mainly Cloud, SonarCloud. And then Cursor I was using a lot, not in the autocomplete function, right? Yeah, to write code. I used more the chat itself. And I stayed there in planning mode, right? Where I gave it a prompt. It plans and then I correct the planning and then it executes. And Cloud is used in the same way too.

And how often do you use these tools?

I use it every day, all day, because that is the recommendation from the people at work, you know? That we use it to make every decision there for now.

And in which activities or types of activity do you usually use AI?

I use it for writing code, right? Product delivery now. Yeah, I also use it for text correction, I still do not use it for creating documentation of things, nor for decision-making, right, even though that is common, I cannot use it yet, but I use it to correct text, correct some theoretical documentation issue that is incorrect, right, with documentation standards. Currently we also have an internal agent at work that helps with review, both PR review, right, in the repository we work in. As well as with the health of the system. We have there the whole system infrastructure, we have internal company agents, they were built by an infrastructure team, which we use to investigate whether a system is up, how the health of the system is, what errors were found. And we are also using it for bug handling, right? We have a rotation both for the team and globally for the whole company, we have this internal AI tool to diagnose a bug, right? Whether there was a problem there with the call, whether it is a bug or whether it is some common client error.

And can you describe a recent case in which you used AI to perform some activity?

Recently we needed to deliver a demand at work, for example, for an architectural decision that would have been implemented in the system. We did the whole process of architectural decision-making in a call. Then we used it, we the team. We used AI both for transcreation of the call where the meetings happen, right? as well as for processing the meeting transcript to break it into cards and break it into documentation. We were discussing everything that would be done and then we were using AI to do that. Yeah, that is the standard they are already using inside there. And then, yeah, then for execution of the cards, right, from the Sprint, all the Sprint cards from the teams were coordinated through, we created skills, skills in Cloud in Anthropic there, they are kind of functionalities inside the system there, of the AI, of the chat, right? That you can create for specific tasks. Then recently I had created a skill in Cloud for the team, where they only pass the context and what is to be done and then it builds a proper prompt for the AI planning, right? Following the prompt engineering techniques there, building the Chain-of-Thought, building context. And then I used AI to write this skill too, right? Which is almost like a markdown, like an algorithm, only very high level, right? Like, if you receive text in this format, transform it into this format, etc. And then the delivery of these cards was also all done with AI, where we used this prompt engineering skill to build the action plan to execute the code, right? To build and write the code itself. And then, yeah, we take there the whole description of the card that was previously made by AI and we build the planning. Then the planning will be the step, again, a very high-level algorithm that the AI will answer you with, what has to be done, which file it will modify, how it will

modify it, what the code will look like before, how the code was before, how it will look after. Then if I have any adjustment to make, I make it there in the planning and if not I tell it to execute. And then, like, I also cannot use AI for code review, but I used it to review the code. But I did not really like the review and had to review all the code of the other people who worked on the cards again, right?

And after you have the meeting, and it builds the cards, do you review these cards or or leave them there as it made them?

Yeah, no. We do not review the cards and that even caused a big problem in this sprint, yeah, that I was mentioning. Because it disregarded some priorities, right? Like, some cards depended on others and some actions should have been taken before others. And then the AI was not able to detect this priority and order, right, of task execution. And we ended up having to, after having implemented a lot of things, we had a problem there and we had to think like, folks, there is something wrong here because this card here is mentioning things that this card here is also mentioning. Would it not have been better to have done this before? What is this? Then yes, we had to review what it had done, correct it, create one more card too, that was the case there, so that we could follow properly with the end of the sprints there.

And these documents that it creates, the cards, the skills, everything else, is it all markdown? It is all markdown, basically it is all markdown.

And then do you pass these markdowns to it to use as context to generate the other things or not? How does it work?

Yeah, we have there the repository of the whole system, right? With all the system code. And then every time we create a skill, some documentation, we generate there a markdown that is saved inside the repository. Then, every time we are going to interact with it, with the chat, these Markdown files become commands inside the chat. According to the documentation, okay, then I put the name of the documentation which is the name of the Markdown file and then I call it inside the chat, you know? And then it will already understand that the Markdown I am talking about is the information that I am giving that Markdown, you know?

So, you do the planning, create the cards, then generate the skills, right? And then from the skills you do the planning of how it will execute that specific skill and then it generates the code and you just keep reviewing. That's it, that's it, exactly.

And do you keep validating or, like, what it does you agree with?

I keep validating, yeah, I cannot, and work also recommends that we let the AI go freely, right? But I do not really like this way and I still prefer to review step by step there in the planning, like, you know? I think it avoids more, for me it becomes much more confusing to review when everything is ready. And then there are also some things that I keep fixing in the planning, right? But in general, like, as the tool I use is Cloud, Cloud has a default skill of its own there, but that every person who uses Cloud will have, which is, it is saved as a markdown that you call Cloud MD. And then this Cloud MD is where you register kind of your standardizations, many of them for writing code like that. An example that nowadays, people have as normal putting, for example, writing code in AI and the AI putting comments in the code. I do not think this is very nice, because, yeah, I do not see the need to put comments in the code, because a person who is going to take this code, a new person who is going to take this code, will read with with the chat, then they will not read the comment and anyway, I think it becomes a useless text there and I do not like it. And then I save like this, for any planning that I do, for it to follow these rules, do not add comments, do not put, for example, in Go we have a language standard where the names of variables are short like, like the variable is called variable and then the Go standard is that it is called vespa, you know? I also do

not like this type of thing. Then I say there in my Cloud markdown, do not leave atomic variable names. There is this more general review too.

And when you build your planning, when you want to adjust something, do you directly ask it to adjust it? How does your interaction with it work?

The code I use gives you three little options there: if it is correct, then it will execute; if it is not correct; and if it is partially correct, and if it is to edit it puts a text box there, then if there is something incorrect, partially correct, you need to adjust this point, this point, this point following this, this, this, this. Like, I always put what needs to be adjusted and why it needs to be adjusted.

Do you have a PO? How do the requirements for a given task come to you?
We only have a PM.

And then does he participate in this discussion you mentioned that will generate the cards?

Yes. He basically comes with the stories, right? The user stories and then we comment: "Let's do this, this, and this." Then, in terms of product, then basically, there is an hour there of conversation with him about that. And then, once this conversation is closed, we generated, like, we leave this meeting thinking: "Then, we have to do these things, these things, these things to contract this user story, right? Contract this epic. And then after this meeting we have another refinement meeting for the technical part. And then that is when we break it again into this other card.

Ah, so the meeting with the PM does not use AI?

No, no, no. We only use AI to transcribe. Because when we are breaking the card we provide the information from both meetings so that it understands both the product context and the technical context.

And when you use AI in a task, how do you define what needs to be done? You already have this in the meetings, but do you also build it throughout the iterations? Do you keep passing more information to it or is only what you defined there in the cards already enough?

I keep passing more information. Some, for example, I showed something here about authentication, an authentication service here in my system. Yeah, I say, according to such card implement such thing, following the system's business rules which are these, following the system's code standards, this repository, it is the repository, which also, like, we have to pass this extra context beyond the card, because there are some issues inside the company I work for, where we work with a monorepository. Which is a gigantic repository that has several mini repositories, let's say, right? There are several mini repositories inside it. It has to pass all this context in addition to the cards to the AI, so that it does not read the whole project, you know? Also only, for example, that part of the project. And then you implement this thing here for authentication. Then I also give more context. In this project, for example, every new layer of the system, it is a new app, it is a new directory. So, to do this activity, create this new directory, according to such directory that does such thing.

And has the use of AI changed the way you work? If so, how?

Wow, it has changed a lot, yes. Because before I think there were a lot of meetings at work. But there was one day that I would take like, usually there came a day when I had fewer meetings, I would take a day to sit down to write code. It was that very isolated, very focused thing, headphones on, writing, writing, writing. I think nowadays there is not much of that, because writing code is much more high level now. We do not write code itself, we write the prompt that will write the code. But this writing of the code I cannot do in such a closed way, like, putting on headphones and just writing. Because I have a feeling almost like I am asking someone to do something. So, it is almost like an exchange, like that, too. I need to read carefully what it is saying, asking if it is right or not, asking if it can execute this or not. Which I did not have, because I am there in my flow

of thought and it is me and me. Then I feel that with AI it is more of a conversation, like, that I have to analyze more minutely what I am doing, even if this, in the end, gives more work than, for example, manually passing everything that I had to write there in a day, spending a day coding. It is less manual work, but I think it is much more cognitive work, like, of having much more attention, having much more interpretation there of something that previously was very well defined in my head. Because, for example, the AI may suggest a way of implementation that I was not thinking of, but that I find interesting. Or else, it may suggest a way of implementation that is not so interesting. I have there a flow of thought of how the thing will work and these questions and these alternative ways of what I want to implement can work. Yeah, it changes a little the way I work.

When you are doing your work with AI and you want to change something, do you need to go through a prompt or can you change it directly there in the code?

I can change it directly in the code, but I never do that because I am afraid of disturbing the process there. Then I change it by prompt. But there is the possibility, but I change it by prompt.

And what difficulties do you find when using AI?

My difficulties are more a matter of ego. I think it is much more a matter of ego than a monotonous issue. I think manually, technically it is much easier to use. Anyone who goes on the cellphone can use it. But there is a however too, that it also seems that before people would say like, there were studies and evidence and the company itself said that it would be more interesting for you to pass context to the AI model. Then, okay, spends three months saying it, now if you pass less context, it becomes easier for it to understand, you know? There are some things that keep changing and that makes it a little difficult to use for me. And the other thing that is a little difficult for me to use, which is this ego issue, is like: "ah, but I can do this, there is no reason for me to ask it." And then, but life requires a way of producing things faster, even more with AI, now everything seems to be much easier, everything seems to be much faster. So, for me it is difficult to get used to it, at the first moment if they ask me to do something, I still want to go there and try to understand the problem, like, or else do the things manually. And then, I also feel, that I work with this, right? I feel a difficulty sometimes in taking the thought, my idea from my head to the prompt. I have a little more difficulty in this part of explaining how I want things to work. This for me did not work perfectly.

And how was your process of starting to use AI? You said there that there is a certain resistance, right? But did you find it very difficult? How was that?

I found it a little difficult, because the first way I tried to use it, I hired a model that was Cursor, right? The first time I tried to use it was that thing autocompleting what I wrote. But then I did not like that, I thought it disturbed my writing a lot, it disturbed my flow of thought a lot. Then I moved to the chat. Then for the chat, even so, it would write the things and then when I saw that it was getting something wrong, it had already written everything that I had asked. Then I had to go do everything, only in that wrong part there was code that was correct. And then I had to remove the wrong part from the code. And then I moved to planning, which is the way I was most able to adapt and I think it works and I think it is very healthy for work, in terms of security, in terms of putting wrong code in the repository, putting wrong code in production, like, in my philosophy, right, of AI use. And it is what I find most interesting nowadays, it is the way that, for me, became less difficult to use.

And how do you feel using AI today? Do you prefer working with it or do you prefer how it was before?

Man, I do not know if for me there was that much difference. I have been talking with some colleagues, that as I explained now, there is all this difficulty in the cognitive part, which before was a difficulty in the technical part there, right? Of getting tired, of thinking, writing, reviewing, file by file and nowadays we do everything with AI, so this technical part has kind of been made easier.

But there is all this cognitive part there of talking to it, passing what is right and what is wrong. I think it swapped six for half a dozen. Let's put it that way, right? But it remains the same thing, only the difficulty there in the development process for me moved to another stage of the process. And then, I was talking with a colleague of mine once, that this was something common, right? Like, in software development, where people wrote very low-level code, in machine language there, asking the machine. For example, for an addition, what it had to do in memory to add two numbers. And higher-level languages were created that we talk to. I once remembered that I read that about code, Clean Code. And then it says that writing code is nothing more than translating an idea into another language, as, for example, it would be translating a book from English into Portuguese. And then only I am translating a product idea into whatever language. And I think that nowadays we are translating this idea into a very high-level algorithm, so, a step by step, for example, Markdown, which will become code that we already know, which would be C, Java, Go. This C, Java and Go goes to machine language and this machine language will say that, for example, what the computer has to do with the memory. And then, for me that is it, like, we are exchanging one language for another, but like, the work, development continues to be development. I think it did not make anything much easier, because I think it still needs a lot there of human insertion in the writing of the code, you know?

And are there situations in which AI cannot help you?

I think not, like, currently I cannot say no, but like, there are things that, like data analysis, it is something that is a little outside my area, but I need to do a little, right? Development, we have to analyze the data there from, from the system health, from the system infrastructure. Yeah, for example, how many errors happened, how is the health, how is the memory usage, these are data that we have to analyze manually. I am trying to do this with AI, only I still think it is not good enough, there still has not been this transfer of, for example, that I feel with code, that I can kill a lot of manual effort with AI, as in writing code. Data analysis still has a lot of that manual effort of extracting all the data, thinking what I have to do there about that information about the system, right? I think it is not that it cannot help me, but I think it is still not good enough. I think it is still not equivalent, for example, for me, yeah, with code writing.

And do you have any type of practice, guidance that could improve the use of AI? What do you think is still missing for it to improve in software development?

In this part of what could improve in AI, to help me. I do not know how to say very well, like. There are things that I feel could be improved, yes, in artificial intelligence, to help development. But I do not know how to point them out very well, like, you know? But I think that in the use of developers with AI, yeah, I have a lot of interaction with other developers who also use AI to write and I think that with the notion there a little a little of prompt engineering, of how to write a prompt, of how to ask AI to execute something, it was more interesting. And I think that with market trends, I think it will become something that will become much more common, like, right? People are still there very much in the discussion, right? It stays a lot around what vibe coding is and us just passing and the AI spitting out the code. Whether this is ethical or not, whether this is healthy or not, whether this helps the developer or not. I think the conversation had to move a little from there in terms of the market, right? And move much more to like what could improve in daily use is that developers focused on understanding a little more about the functioning for writing code, like. There are many of these problems. It is really like writing code in C. Yeah, the machine will do what you ask it to do, like, you know? If you do not ask properly what has to be done, it will really come out wrong. Then there will be the hallucination problems, the security problems, which I think when you ask properly are much more minimized.